

First week

A social science a government made calculation (managed allocated their resources) unlimited wants corresponding limited resources.

→ There is no economy if the resources are infinite

Free goods: The goods that are available without any effort or money.

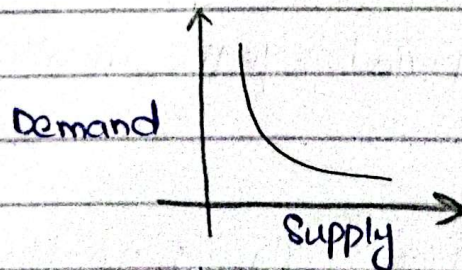
Non-Free goods: The goods that are available only with efforts or money.

Demand and Supply:

If demand is high & supply is low, the price will be high and vice versa.

* Both are inversely proportional to each other and effect the price.

* We can evaluate a project that will be succesful in future can be done by different analysis.



Factors of Production:

There are three factors:

① LAND ② LABOUR ③ CAPITAL STOCK

Interest Rate & Rate of Return

- Time value of Money: It changes with time.
Interest rate is the percentage of interest amount. It will be in percentage & interest will be in dollars.

$I \rightarrow$ Interest (\$, R)

$i \rightarrow$ Interest Rate (%)

∴ you must write the per unit with the answer of interest rate with %.

Terminology and Symbols

$P \rightarrow$ value of money at present time or at time 0

$F \rightarrow$ value of money at some time in Future

$A \rightarrow$ series of consecutive, equal, end, amount of money

$n \rightarrow$ Number of interest period

$i \rightarrow$ Interest rate per time period

$t \rightarrow$ Time, stated in periods; years, months, days

Example 1.3

An employee borrows \$10,000 on May 1, must repay a total of \$10,700 exactly one year later.

Interest Amount and Interest Rate Paid

1) Interest = amount owed now - principal

2) Interest Rate (%) = $\frac{\text{interest occurred per time} \times 100\%}{\text{principal}}$

1) Interest paid = \$10,700 - \$10,000 = \$700

2) Interest rate (%) = $\frac{\$700}{\$10,000} \times 100\% = \boxed{7\% \text{ per year}}$

Example 1.4

Plans to Borrow \$20,000 from a bank for

1 year at 9% interest.

a) compute the interest & total amount due after 1 year

b) Total amount after 1 year, loan interest, rate of 9%.

Total Due = principal (1 + interest rate)

→ From the perspective of a saver, a lender or an investor → Interest Earned

Interest Earned = Total amount now - principal

→ Interest earned over a specific period of time is expressed as a percentage of the original amount and is called rate of Return (ROR)

$$\text{Rate of return (\%)} = \frac{\text{interest occurred per unit time}}{\text{Principal}} \times 100\%$$

Solution:

$$\begin{aligned} \text{Total Due Amount} &= \$20,000 (1 + 9\%) \\ &= \$21,800 \end{aligned}$$

Example #1.5

- calculate the amount deposited 1 year ago to have \$1000 now at an interest rate of 5% per year.
- Calculate amount of interest earned during this time period.

$$\text{Total occurred} = \text{Deposit} + \text{Deposit (interest Rate)}$$

$$\text{Amount owed} = \$10,000$$

$$\text{Interest rate} = 5\%$$

$$\text{Total amount occurred} = \text{Principal} (1 + i)$$

$$\$1000 = \text{Principal} (1 + 5\%)$$

$$\frac{\$1000}{1.05} = \text{Principal}$$

$$\text{Principal} = \$952.38$$

Amount of interest Earned

$$\text{Interest Earned} = \text{Total amount now} - \text{Principal}$$

$$\text{Interest Earned} = \$1000 - \$952.38$$

$$= \$47.62$$

Example 1.6

Borrowed \$5000, Two ways loan repay
Engineering Economics symbol?

- a) Five equal annual installments with interest based on 5% per year

$$P = \$5000, i = 5\% \text{ per year}$$

$$n = 5 \text{ years}, A = ?$$

P = value of money at a time designated
Present worth (P)

i = Interest Rate per time period

n = Number of interest periods

A = Annual worth $A(W)$

- b) One Payment 3 years from now with interest based 7% per year

$$P = \$5000, i = 7\%, n = 3 \text{ years}, F = ?$$

F = value of money at some future time

Example 1.7

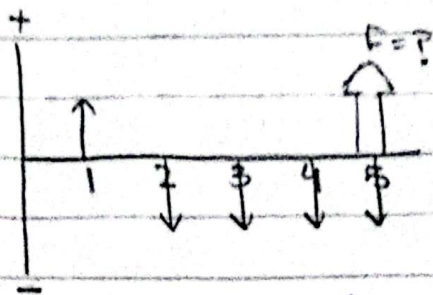
Statement From Book - Data is given below

$$P = \$5000, i = 6\% \text{ per year}, A = \$1000 \text{ for years}$$

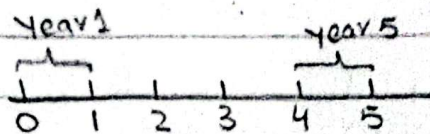
$$n = 5, F = \text{at end of 6th year}$$

• CASH FLOWS: ESTIMATION & DIAGRAMMING

Net cash flow = cash inflows - cash outflows



example of +ve and -ve cash flows



A typical cash flow scale of 5 Years

+ve axis : Getting money (withdrawal)

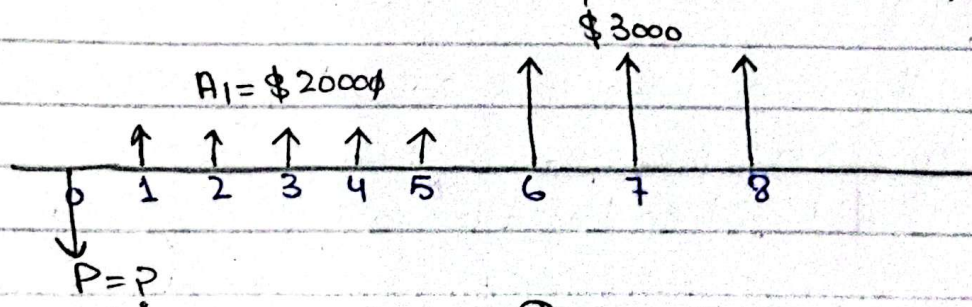
-ve axis : Paying money (deposit)

length of line shows amount of money

Example 1.10

Data: $P=?$, $A_1 = \$2000$, $A_2 = \$3000$

$n_1 = 5$ years, $n_2 = 3$ years, $i = 8.5\%$ per year



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Simple and Compound Interest

- Based on principal amount is simple interest
- calculated on principal amount & outstanding amounts leads to compound interest.

$$\ast \text{ Simple interest} = (\text{Principal})(n)(i)$$

$$\ast \text{ Compound interest} = (\text{Principal} + \text{all accurate interest})(\text{interest rate})$$

Example 1.14

Lended amount = \$100,000; 3 years loan
10% per year; money repay end of 3 years

$$\text{Interest per year} = (100,000)(10\%)$$

$$= 10000$$

$$\text{Total interest} = (100,000)(3)(0.10)$$

$$= 30000$$

$$\text{Total due} = 100,000 + 30000 = \$130000$$

Now after 3 years

$$\text{Interest per year} = (100,000)(0.10) = 10000$$

$$1 \text{ year} = 100000 + 10000 = 110000$$

$$2 \text{ year} = 110,000 + 11000 = 121,000$$

$$3 \text{ year} = (121)(0.10) = 12,100$$

$$121,000 + 12,100 = \$133,100$$

Table 1.1 Repay \$5000 in 5 years at 8% year

(simple interest)

End of year	Interest owed for year	Total owed at end of year	End of year payment	Total owed after payment
0		\$5000		
1	400	\$5400	0	\$5400
2	400	\$5800	0	\$5800
3	400	\$6200	0	\$6200
4	400	\$6600	0	\$6600
5	400	\$7000	\$7000	0

b) Repay \$5000 in 5 years at 8% (Compound)

0		\$5000		
1	400	\$5400	0	\$5400
2	432 ^{$5400 \times \frac{8}{100}$}	\$5832 ^{$5400 + 432$}	0	\$5832
3	466.56	\$6298.56	0	\$6298.56
4	503.88	\$6802.44	0	\$6802.44
5	544.19	\$7390.19	\$7346.63	0
		\$7346.63		

c) Pay annually (Compound)

0		\$5000		
1	400	\$5400	400	\$5000
2	400	\$5400	400	\$5000
3	400	\$5400	400	\$5000
4	400	\$5400	400	\$5000
5	400	\$5400	5400	0

d) 1/5th of principal amount annually

0		\$5000		
1	400	\$5400	\$1600	\$6000
2	320	\$4320	\$1320	\$3000
3	240	\$3240	\$1160 ^{\$1240}	\$2000
4	160	\$2160	\$1080 ^{\$1160}	\$1000
5	80	\$1080	\$1080	0

$$\begin{aligned}
 * A &= P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] \\
 &= 5000 \left[\frac{0.08(1+0.08)^5}{(1+0.08)^5 - 1} \right] \\
 &= \$1252.28
 \end{aligned}$$

0		\$5000		
1	400	\$5400	\$1252.28	\$4147.72
2	331.81	4479.53	1252.28	\$3227.25
3	258.18	3485.43	1252.28	\$2233.15
4	178.65	2411.802	1252.28	\$1159.52
5	92.76	1252.28	1252.28	0

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Minimum Attractive Rate of Return (MARR)

Financing
 ↳ Equity ↳ Debt

Chapter no 2

Single amount factor (P/F, F/P)

$$F = P(1+i)^n \text{ or } P = \frac{F}{(1+i)^n}$$

P has value than F
 less

there are three methods to solve these questions

- ↳ Using tables
- ↳ Using formula
- ↳ Using MS Excel

Format to attempt the question :-
Four Steps

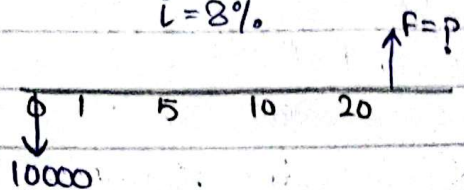
- 1- Draw CashFlow diagram
- 2- Write equation i.e. $F = P(F/P, i, n)$
- 3- Write formula
- 4- Put values and write answer

Example 2.1

Bonus = \$10,000 ; Profit = 8% per year ,

After 20 years

$P = \$10,000$, $F = P$, $i = 8\%$, $n = 20$ years
 $i = 8\%$



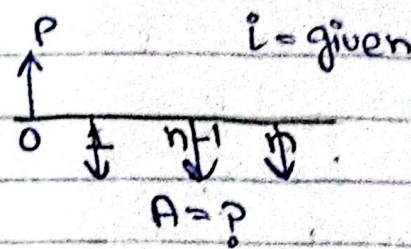
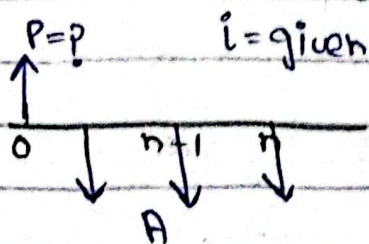
$$\begin{aligned} \rightarrow F &= P(F/P, i, n) \\ &= 10000(F/P, 0.08, 20) \end{aligned}$$

$$\begin{aligned} \rightarrow F &= P(1+i)^n = 10000(1+0.08)^{20} \\ &\boxed{F = \$46609.57} \end{aligned}$$

2nd Factor: (P/A or A/P) (Find/given)
Present worth and Capital Recovery

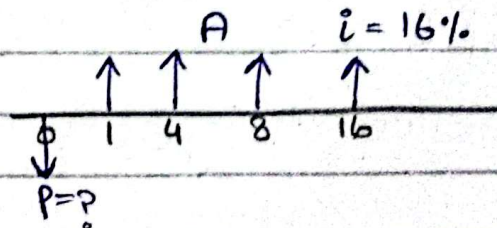
$$A = P \left[\frac{i(1+i)^n}{(1+i)^n - 1} \right] \quad \text{and}$$

$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$



Example 2.3

$A = \$600$, $n = 9$, $i = 16\%$, $P = ?$



$$P = \$600(P/A, 0.16, 9)$$

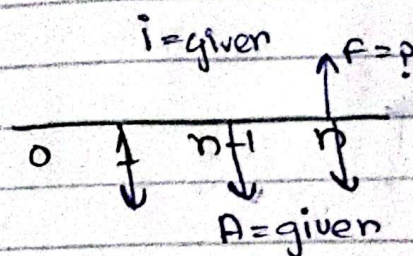
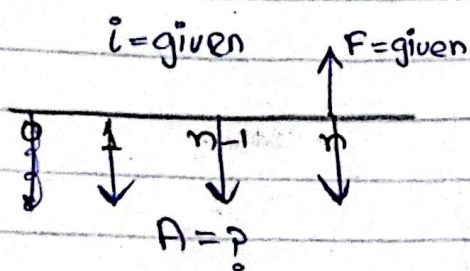
$$P = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] = (600) \left[\frac{(1+0.16)^9 - 1}{(0.16)(1+0.16)^9} \right]$$

$$\boxed{P = \$2763.92}$$

3rd Factor: (F|A or A|F)

$$A = F \left[\frac{i}{(1+i)^n - 1} \right] \text{ and}$$

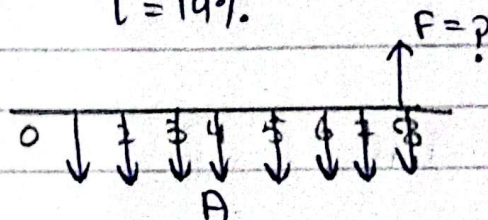
$$F = A \left[\frac{(1+i)^n - 1}{i} \right]$$



Example 2.5

$A = \$1,000,000$, $n = 8$, $i = 14\%$
starting 1 year

$i = 14\%$



$$F = 1,000,000 (F|A, 0.14, 8)$$

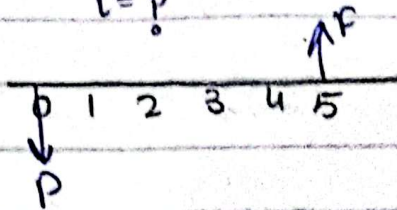
$$F = A \left[\frac{(1+i)^n - 1}{i} \right] = [1,000,000] \left[\frac{(1+0.14)^8 - 1}{0.14} \right]$$

$$F = \$13.2 \text{ million}$$

Example:

$$P = 30,000, n = 5, F = \$50,000, i = ?$$

$$i = ?$$



$$F = P(F/P, i, n)$$

$$50000 = 30000(F/P, i, 5)$$

$$F = P(1+i)^n$$

$$\frac{F}{P} = (1+i)^n$$

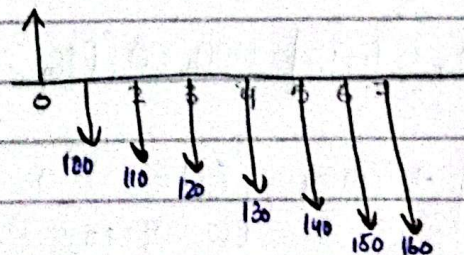
$$\sqrt[n]{\frac{F}{P}} = 1+i$$

$$i = \sqrt[n]{\frac{F}{P}} - 1 = \sqrt[5]{\frac{50000}{30000}} - 1$$

$$i = 10.76\%$$

⑤

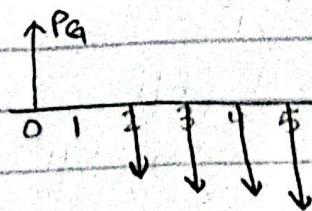
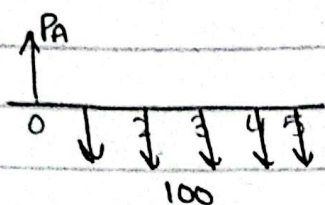
Arithmetic Gradient (A/G, P/G)



$$P_T = (100)(P/F, 10\%, 1) + 110(P/F, 10\%, 2) + 120(P/F, 10\%, 3) \dots$$

$$\rightarrow P_G = \frac{G}{i} \left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n} \right]$$

$$\rightarrow A_G = G \left[\frac{1}{i} - \frac{n}{(1+i)^n - 1} \right] \quad \therefore \text{gradient} = 10$$

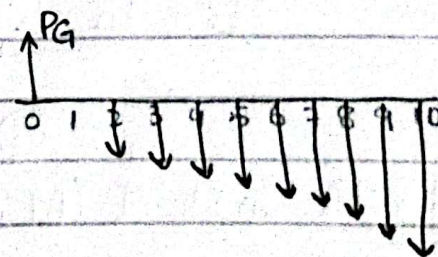
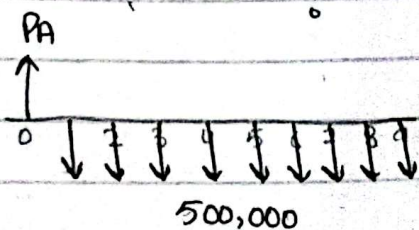


$$P_T = P_A + P_G$$

$$P_A = 100 (P/A, 10\%, 5)$$

$$P_G = 10 (P/G, 10\%, 5)$$

Example: Deposit \$500,000 next year, \$100,000 increase for 9 years - Calculate present worth = P , $i = 5\%$.



$$P_T = P_A + P_G$$

$$= 500,000 [P/A, 5\%, 10] + 100,000 (P/G, 5\%, 10)$$

$$= 500,000 \left[\frac{(1+0.05)^{10} - 1}{(0.05)(1+0.05)^{10}} \right] + \frac{100,000}{5} \left[\frac{(1+0.05)^{10} - 1}{(0.05)(1+0.05)^{10}} - \frac{10}{(1+0.05)^{10}} \right]$$

$$= \$7026.05$$

$$\rightarrow AG = G \left[\frac{1}{i} - \frac{n}{(1+i)^n - 1} \right]$$

$$AT = AA + AG$$

$\therefore$ It is same annual to annual

(6)

$\therefore$ series gradually increases by fixed value is Arithmetic series.

$\rightarrow$ It increases Linearly.

Geometric series:

Increasing by a fixed percentage.

$\rightarrow$ It increases exponentially.

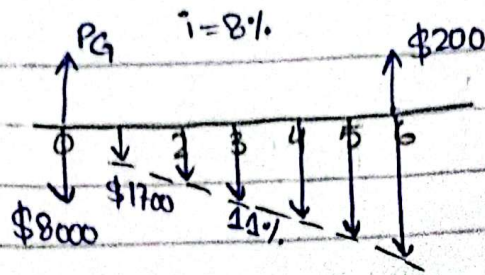
$$\Rightarrow Pg = A(PIA, g, i, n)$$

$$(PIA, g, i, n) = \begin{cases} \frac{1 - \left(\frac{1+g}{1+i}\right)^n}{i - g} & i \neq g \\ \frac{n}{1+i} & i = g \end{cases}$$

i and g will be decimals.

Example 2.11

cost = \$8000 ; $n = 6$ years ; \$200 salvage value
\$1700 in 1st year ; $g = 11\%$; $i = 8\%$.



$$= \frac{1 - \left(\frac{1+0.11}{1+0.8} \right)^6}{0.8 - 0.11}$$

$$P = -8000 - 1700(P/A, 11\%, 8\%, 6) + 200(P/F, 8\%, 6)$$

$$= -8000 - 1700 \left[\frac{1 - \left(\frac{1+0.11}{1+0.8} \right)^6}{0.8 - 0.11} \right] + \frac{200}{(1.08)^6}$$

$$= \$+17,999$$



Chapter no 3

Combining Factors